

Important Questions for Class 12

Maths

Chapter 11- Three Dimensional Geometry

Very Short Answer Questions

1 Mark

1. What is the distance of point (a,b,c) from x-axis?

Ans: The distance of point (a,b,c) from x-axis is equal to a.

2. What is the angle between $2x = 3y = -z$ and $6x = -y = -4z$?

Ans: Let angle between $2x = 3y = -z$ and $6x = -y = -4z$ is θ .

$$2x = 3y = -z \text{ can be written as } \frac{x}{3} = \frac{y}{2} = \frac{-z}{6}$$

$$6x = -y = -4z \text{ can be written as } \frac{x}{2} = \frac{-y}{12} = \frac{-z}{3}$$

$$\cos \theta = \frac{(3)(2) + (2)(-12) + (-6)(-3)}{\sqrt{(3)^2 + (2)^2 + (-6)^2} \sqrt{(2)^2 + (12)^2 + (3)^2}}$$

$$\Rightarrow \cos \theta = \frac{-36}{7\sqrt{157}}$$

$$\Rightarrow \theta = \cos^{-1} \left(\frac{-36}{7\sqrt{157}} \right)$$

3. Write the equation of line passing through (2,3,-5) and parallel to line

$$\frac{x-2}{3} = \frac{y-4}{3} = \frac{z+5}{-1} ?$$

Ans: Directional ratios of the required line are (3,4,-1).

This line passes through (2,3,-5).

So, the required line equation is $\frac{x-2}{3} = \frac{y-4}{3} = \frac{z+5}{-1}$.

4. Write the equation of a line passing through (1,1,-3) and perpendicular

to $\vec{r} \cdot (\vec{i} - \vec{j} + 3\vec{k}) = 5$?

Ans: Directional ratios of the required line are $(1, 1, -3)$.

This line passes through $(1, 2, 3)$.

So, the required line equation is $\frac{x-1}{1} = \frac{y-2}{1} = \frac{z+3}{-3}$.

5. What is the value of λ for which the lines $\frac{x-1}{2} = \frac{y-3}{5} = \frac{z-1}{\lambda}$ and

$\frac{x-2}{3} = \frac{y+1}{-2} = \frac{z}{2}$ are perpendicular to each other?

Ans: Given two lines are perpendicular.

So,

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

$$\Rightarrow (2)(3) + (5)(-2) + (\lambda)(2) = 0$$

$$\Rightarrow \lambda = 2$$

6. If a line makes angles α, β, γ with the coordinate axes, then the value of $\sin^2\alpha + \sin^2\beta + \sin^2\gamma$?

Ans: We know that if a line makes angles α, β, γ with the coordinate axes, then the value of

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$\Rightarrow 3 - (\sin^2\alpha + \sin^2\beta + \sin^2\gamma) = 1$$

$$\Rightarrow \sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$$

7. Write the $\vec{r} = (\vec{i} - \vec{j}) + \lambda(2\hat{j} - \hat{k})$ into cartesian form? Ans:

Ans: Given $\vec{r} = (\vec{i} - \vec{j}) + \lambda(2\hat{j} - \hat{k})$

The cartesian form of $\vec{r} = (\vec{i} - \vec{j}) + \lambda(2\hat{j} - \hat{k})$ is $\frac{x-1}{2} = \frac{y+1}{0} = \frac{z-0}{-1}$.

8. If the direction ratios of a line are 1, -2, 2 then what are the direction cosines of the line?

Ans: If direction ratios are a, b, c , then the directional cosines are

$$\frac{a}{\sqrt{a^2 + b^2 + c^2}}, \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \frac{c}{\sqrt{a^2 + b^2 + c^2}}.$$

So, the directional cosines are

$$\frac{1}{\sqrt{1^2 + (-2)^2 + 2^2}}, \frac{-2}{\sqrt{1^2 + (-2)^2 + 2^2}}, \frac{2}{\sqrt{1^2 + (-2)^2 + 2^2}}$$

$$\Rightarrow \frac{1}{3}, \frac{-2}{3}, \frac{2}{3}$$

9. Find the angle between the planes $2x - 3y + 6z = 9$ and xy - plane?

Ans: Equation of XY plane is $z = 0$.

Angle between $z = 0$ and $2x - 3y + 6z = 9$ is

$$\theta = \cos^{-1} \left(\frac{2(0) - 3(0) + 6(1)}{\sqrt{4 + 9 + 36} \sqrt{1}} \right)$$

$$\Rightarrow \theta = \cos^{-1} \left(\frac{6}{7} \right)$$

10. Write equation of a line passing through $(0,1,2)$ and equally inclined to coordinate axes.

Ans: Given that the line is equally inclined with all axes.

So, we consider the directional ratios as $1,1,1$.

This line passes through $(0,1,2)$.

The line equation is $x - 0 = y - 1 = z - 2$.

11. What is the perpendicular distance of plane $2x - y + 3z = 10$ from origin?

Ans: The perpendicular distance from $2x - y + 3z = 10$ from origin is

$$d = \left| \frac{10}{\sqrt{2^2 + 1^2 + 3^2}} \right| = \frac{10}{\sqrt{14}}$$

12. What is the y -intercept of the plane $x - 5y + 7z = 10$?

Ans: The y -intercept of the plane $x - 5y + 7z = 10$ is equal to 5.

13. What is the distance between the planes $2x + 2y - z + 2 = 0$ and

$$4x + 4y - 2z + 5 = 0?$$

Ans: The given planes are $2x + 2y - z + 2 = 0$ and $4x + 4y - 2z + 5 = 0$.

$$4x + 4y - 2z + 5 = 0 \text{ can be written as } 2x + 2y - z + \frac{5}{2} = 0.$$

So these two planes are parallel.

$$\text{The distance between these two planes is } \frac{\left|2 - \frac{5}{2}\right|}{\sqrt{2^2 + 2^2 + 1^2}} = \frac{1}{6}$$

14. What is the distance between the planes which cuts off equal intercepts of unit length on the coordinate axes?

Ans: The equation of the plane whose intercepts are (a, b, c) is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

We are given that the intercepts are equal and their value is equal to 1.

So, the new equation of the plane is $x + y + z = 1$.

15. Are the planes $x + y - 2z + 4 = 0$ and $3x + 3y - 6z + 5 = 0$ intersecting?

Ans: The given planes are $x + y - 2z + 4 = 0$ and $3x + 3y - 6z + 5 = 0$.

$$3x + 3y - 6z + 5 = 0 \text{ can be written as } x + y - z + \frac{5}{3} = 0.$$

So, these two planes are parallel and they will not intersect.

16. What is the equation of the plane through the point $(1, 4, -2)$ and parallel to the plane $-2x + y - 3z = 7$?

Ans: The equation of the plane parallel to $ax + by + cz + d = 0$ is

$$ax + by + cz + k = 0.$$

So, the equation of the plane parallel to $-2x + y - 3z = 7$ is $-2x + y - 3z + k = 0$

Now place $(1, 4, -2)$ in $-2x + y - 3z + k = 0$.

$$-2 + 8 + 6 + k = 0$$

$$\Rightarrow k = -12$$

17. Write the vector equation which is at a distance of 8 units from the origin and is normal to the vector $2\hat{i} + \hat{j} + 2\hat{k}$?

Ans: The normal to the plane is $2\hat{i} + \hat{j} + 2\hat{k}$.

So, the equation of the plane is $2x + y + 2z + d = 0$.

The distance of the plane $ax + by + cz = d$ to the origin is $\left| \frac{d}{\sqrt{a^2 + b^2 + c^2}} \right|$.

So,

$$\left| \frac{d}{\sqrt{2^2 + 1^2 + 2^2}} \right| = 8$$

$$\Rightarrow d = \pm 24$$

Hence, the equation of the plane is $2x + y + 2z \pm 24 = 0$.

18. What is equation of the plane if the foot of perpendicular from origin to this plane is (2,3,4)?

Ans: The direction ratios of the normal is $(2 - 0, 3 - 0, 4 - 0) = (2, 3, 4)$.

So, the equation of the plane is $2x + 3y + 4z + d = 0$.

As (2,3,4) lies on the plane $2x + 3y + 4z + d = 0$,

$$2(2) + 3(3) + 4(4) + d = 0$$

$$\Rightarrow d = -29$$

So, the equation of the plane is

$$2x + 3y + 4z - 29 = 0$$

19. Find the angles between the planes $\hat{r} \cdot (\hat{i} - 2\hat{j} - 2\hat{k}) = 1$ and

$$\hat{r} \cdot (3\hat{i} - 6\hat{j} + 2\hat{k}) = 0$$

Ans: The angle between $\hat{r} \cdot (\hat{i} - 2\hat{j} - 2\hat{k}) = 1$ and $\hat{r} \cdot (3\hat{i} - 6\hat{j} + 2\hat{k}) = 0$ is

$$\theta = \cos^{-1} \left(\frac{1(3) + (-2)(-6) + (-2)(2)}{\sqrt{1^2 + 2^2 + 2^2} \sqrt{3^2 + 6^2 + 2^2}} \right) = \cos^{-1} \left(\frac{11}{21} \right)$$

20. What is the angle between the line $\frac{x+1}{3} = \frac{2y-1}{4} = \frac{2-z}{-4}$ and the line plane $2x + y - 2z = 4$?

Ans: The directional ratios of the line is (3,4,-4).

The directional ratios of the normal of the plane is (3,-6,2).

Let the angle between them is θ .

$$\theta = \cos^{-1} \left(\frac{3(3) + (4)(-6) + (-4)(2)}{\sqrt{3^2 + 4^2 + 4^2} \sqrt{3^2 + 6^2 + 2^2}} \right) = \cos^{-1} \left(\frac{-23}{7\sqrt{41}} \right)$$

21. If O is origin the value of OP is 3 with direction ratios proportional to -1,2,3 then what are the coordinates of P?

Ans: The equation of the line OP is $\frac{x}{-1} = \frac{y}{2} = \frac{z}{-2}$.

$$\text{Let } \frac{x}{-1} = \frac{y}{2} = \frac{z}{-2} = k$$

$$\Rightarrow x = -k, y = 2k, z = -2k$$

The distance OP is equal to 3.

So,

$$\sqrt{k^2 + 4k^2 + 4k^2} = 3$$

$$\Rightarrow k = \pm 1$$

The coordinates of P are $(\mp 1, \pm 2, \mp 2)$

22. What is the distance between the line $\vec{r} = 2\hat{i} - 2\hat{j} + 3\hat{k} + \lambda(\hat{i} + \hat{j} + 4\hat{k})$ from the plane $\vec{r} \cdot (-\hat{i} + 5\hat{j} - \hat{k}) + 5 = 0$?

Ans: If the normal of the plane and the given vector are perpendicular, then the plane and given vector are said to be parallel.

Let us assume the angle between the normal and given vector is θ .

$$\cos \theta = \left| \frac{1(-1) + 1(5) + 4(-1)}{\sqrt{1^2 + 1^2 + 4^2} \sqrt{1^2 + 5^2 + 1^2}} \right| = 0$$

So, it is clear that the plane and the given vector are parallel.

As the distance between two parallel line is constant, let us consider a point on the line and we will find the distance between the point and the plane.

Let us assume the distance is equal to d.

$$d = \left| \frac{5}{\sqrt{1^2 + 1^2 + 4^2} \sqrt{1^2 + 5^2 + 1^2}} \right| = \frac{5}{9\sqrt{6}}$$

23. Write the line $2x = 3y = 4z$ in vector form?

Ans: Given line $2x = 3y = 4z$

The given line can be written as $\frac{x}{\frac{1}{2}} = \frac{y}{\frac{1}{3}} = \frac{z}{\frac{1}{4}}$.

The vector for this line is $\lambda \left(\frac{1}{2}\hat{i} + \frac{1}{3}\hat{j} + \frac{1}{4}\hat{k} \right)$.

Short Answer Questions

4 Marks

24. The line $\frac{x-4}{1} = \frac{2y-4}{2} = \frac{k-z}{-2} = a$ lies exactly in the plane $2x - 4y + z = 7$.

Find the value of k.

Ans: Let $\frac{x-4}{1} = \frac{2y-4}{2} = \frac{k-z}{-2} = a$

As $\frac{x-4}{1} = \frac{2y-4}{2} = \frac{k-z}{-2} = a$ lies exactly in the plane $2x - 4y + z = 7$, place the value of a is equal to 0.

$$\Rightarrow x = 4, y = 2, z = k$$

Place $(4, 2, k)$ in $2x - 4y + z = 7$

$$2(4) - 4(2) + k = 7$$

$$\Rightarrow k = 7$$

So, the value of k is equal to 7.

25. Find the equation of the plane containing the points

$(0, -1, -1), (-4, 4, 4), (4, 5, 1)$. Also show that $(3, 9, 4)$ lies on the plane.

Ans: The equation of the plane containing the points

$(0, -1, -1), (-4, 4, 4), (4, 5, 1)$ is

$$\begin{vmatrix} x-0 & y+1 & z+1 \\ -4 & 4 & 4 \\ 4 & 5 & 1 \end{vmatrix} = 0$$

$$\Rightarrow 4x - 5y + 9z + 3 = 0$$

Place $(3, 9, 4)$ in $4x - 5y + 9z + 3 = 0$.

$$\Rightarrow 4(3) - 5(9) + 9(4) - 3 = 0$$

So, it is clear that $(3, 9, 4)$ lies on equation of the plane containing the points $(0, -1, -1), (-4, 4, 4), (4, 5, 1)$

26. Find the equation of the plane which is perpendicular to the plane $\vec{r} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 4$ and $\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) + 5 = 0$ and which is containing the line of intersection of the planes $\vec{r} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 4$ and $\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) + 5 = 0$.

Ans: The required planes pass through intersection of $\vec{r} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 4$ and $\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) + 5 = 0$ is

$$\vec{r} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) - 4 + \lambda (\vec{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) + 5) = 0$$

$$\Rightarrow \vec{r} \cdot ((2\lambda + 1)\hat{i} + (\lambda + 2)\hat{j} + (3 - \lambda)\hat{k}) + (5\lambda - 4) = 0$$

This plane is perpendicular to $\vec{r} \cdot (5\hat{i} + 3\hat{j} + 6\hat{k}) + 8 = 0$.

$$5(2\lambda + 1) + 3(\lambda + 2) + 6(3 - \lambda) = 0$$

$$\Rightarrow \lambda = \frac{7}{19}$$

$$\text{Place } \lambda = \frac{7}{19} \text{ in } \vec{r} \cdot ((2\lambda + 1)\hat{i} + (\lambda + 2)\hat{j} + (3 - \lambda)\hat{k}) + (5\lambda - 4) = 0.$$

$$\text{The final equation is } \vec{r} \cdot (33\hat{i} + 45\hat{j} + 50\hat{k}) - 41 = 0$$

27. If l_1, m_1, n_1 and l_2, m_2, n_2 are direction cosines of two mutually perpendicular lines, show that the direction cosines of line perpendicular to both of them are

$$m_1n_2 - n_1m_2, n_1l_2 - l_1n_2, l_1m_2 - m_1l_2$$

Ans: The vector equation of the perpendicular line is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = (m_1n_2 - n_1m_2)\hat{i} + (n_1l_2 - l_1n_2)\hat{j} + (l_1m_2 - m_1l_2)\hat{k}$$

So, the directional cosines are $(m_1n_2 - n_1m_2)\hat{i}, (n_1l_2 - l_1n_2)\hat{j}, (l_1m_2 - m_1l_2)\hat{k}$.

28. Find vector and Cartesian equation of a line passing through a point with position vectors $2\hat{i} + \hat{j} + 2\hat{k}$ and which is parallel to the line joining the points with position vectors $-\hat{i} + 4\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + 2\hat{k}$

Ans: Line perpendicular to $-\hat{i} + 4\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + 2\hat{k}$ is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 4 & 1 \\ 1 & 2 & 2 \end{vmatrix} = 6\hat{i} + 3\hat{j} - 6\hat{k}.$$

Equation of the line passing through $2\hat{i} + \hat{j} + 2\hat{k}$ and parallel to $6\hat{i} + 3\hat{j} - 6\hat{k}$ is

$$\frac{x-2}{2} = \frac{y-1}{1} = \frac{z-2}{-2}.$$

Its vector form is $\vec{r} = (2\vec{i} + \vec{j} + 2\vec{k}) + \lambda(6\vec{i} + 3\vec{j} - 6\vec{k})$

29. Find the equation of the plane passing through the point $(3,4,2)$ and $(7,0,6)$ and is perpendicular to the plane $2x - 5y - 15 = 0$.

Ans: Let us assume the equation of the plane is $ax + by + cz = d$.

$ax + by + cz = d$ is perpendicular to $2x - 5y - 15 = 0$.

$$\Rightarrow 2a - 5b = 0$$

$$\Rightarrow a = \frac{5b}{2} \dots (1)$$

$ax + by + cz = d$ passes through $(3,4,2)$

$$3a + 4b + 2c = d \dots (2)$$

$ax + by + cz = d$ passes through $(7,0,6)$

$$7a + 6c = d \dots (3)$$

Place equation (1) in equation (2)

$$\frac{23b}{2} + 2c = d \dots (4)$$

Place equation (3) in equation (2)

$$\frac{35b}{2} + 6c = d \dots (5)$$

From equation (4) and equation (5), then

$$b = \frac{2c}{3} \dots (6)$$

Now place equation (1), (6) in equation (2).

$$\frac{5b}{2} + b + \frac{3b}{2} = d$$

$$\Rightarrow d = 5b \dots (7)$$

Place equation (1), (6) and (7) in $ax + by + cz = d$, we get

$$5x + 2y + 3z = 10$$

So, $5x + 2y + 3z = 10$ is the required plane.

30. Find equation of the plane through line of intersection of planes $\hat{r} \cdot (2\hat{i} + 6\hat{j}) + 12 = 0$ and $\hat{r} \cdot (3\hat{i} - \hat{j} + 4\hat{k}) = 0$ which is at a unit distance from origin.

Ans: Let us consider $ax + by + cz = d$.

Cartesian form of $\hat{r} \cdot (2\hat{i} + 6\hat{j}) + 12 = 0$ is $x + 3y + 6 = 0$.

Cartesian form of $\hat{r} \cdot (3\hat{i} - \hat{j} + 4\hat{k}) = 0$ is $3x - y + 4z = 0$.

Direction ratios of the plane perpendicular to the planes is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 0 \\ 3 & -1 & 4 \end{vmatrix} = 12\hat{i} - 4\hat{j} - 10\hat{k}$$

The intersection line of $x + 3y + 6 = 0$ and $3x - y + 4z = 0$ is

$$\frac{x}{1} = \frac{3y + 6}{-1} = \frac{-(6z + 3)}{5}$$

So, we can say that $(0, -2, -2)$ lies on the required plane $ax + by + cz = d$.

$$12(0) - 4(-2) - 10(-2) = d$$

$$\Rightarrow d = 28$$

So, the final equation is

$$12x - 4y - 10z = 28$$

$$\Rightarrow 6x - 2y - 5z = 14$$

31. Find the image of the point $P(3, -2, 1)$ in the plane $3x - y + 4z = 2$

Ans: Assume Q be the image of the point $P(3, -2, 1)$ in the plane

$$3x - y + 4z = 2$$

So, PQ is normal to the plane.

So, the direction ratios of PQ are 3, -1 and 4.

Equation of line PQ is $\frac{x-3}{3} = \frac{y+2}{-1} = \frac{z-1}{4} = k$.

So, we can write point Q as $Q(3k+3, -k-2, 4k+1)$.

The midpoint of PQ lies on $3x - y + 4z = 2$.

Midpoint of PQ is $\left(\frac{3k+6}{2}, \frac{-k-4}{2}, 2k+1\right)$.

$$3\left(\frac{3k+6}{2}\right) - \left(\frac{-k-4}{2}\right) + 4(2k+1) = 2$$

$$\Rightarrow k = -1$$

So, the image is $Q(0, -1, -3)$

32. Find the equation of a line passing through $(2, 0, 5)$ and which is parallel to line $6x - 2 = 3y + 1 = 2z - 2$?

Ans: The equation of the line $6x - 2 = 3y + 1 = 2z - 2$ can be written as

$$\frac{x - \frac{1}{3}}{\frac{1}{6}} = \frac{y + \frac{1}{3}}{\frac{1}{3}} = \frac{z - 1}{\frac{1}{2}}$$

The equation of the required plane can be written as $\frac{x}{6} + \frac{y}{3} + \frac{z}{2} = d$.

This plane $\frac{x}{6} + \frac{y}{3} + \frac{z}{2} = d$ passes through $(2, 0, 5)$.

$$\Rightarrow d = \frac{2}{6} + 0 + \frac{5}{2}$$

$$\Rightarrow d = \frac{17}{6}$$

So, the equation of the plane is $\frac{x}{6} + \frac{y}{3} + \frac{z}{2} = \frac{17}{6}$.

33. Find equations of a plane passing through the points $(2, -1, 0)$ and $(3, -4, 5)$ and parallel to the line $2x = 3y = 4z$.

Ans: The equation of the line $2x = 3y = 4z$ can be written as

$$\frac{x}{\frac{1}{2}} = \frac{y}{\frac{1}{3}} = \frac{z}{\frac{1}{4}}$$

The equation of the required plane can be written as $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = d$.

This plane $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = d$ passes through $(2, -1, 0)$.

$$\Rightarrow d = \frac{2}{2} - \frac{1}{3}$$

$$\Rightarrow d = \frac{2}{3}$$

So, the equation of the plane is $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = \frac{2}{3}$.

34. Find equations of a plane passing through the points of intersection of line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane $x - y + z = 5$?

Ans: The directional ratios of the required line are $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 12 \\ 1 & -1 & 1 \end{vmatrix} = 16\hat{i} + 9\hat{j} - 7\hat{k}$.

$$\text{Let } \frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12} = k$$

So,

$$x = 3k + 2$$

$$y = 4k - 1$$

$$z = 12k + 2$$

So, the equation of the plane is $16x + 9y - 7z + d = 0$

Place this in the plane $16x + 9y - 7z + d = 0$

$$48k + 16 + 36k - 9 - 84k - 14 + d = 0$$

$$\Rightarrow d = 21$$

So, the equation of the plane is $16x + 9y - 7z + 21 = 0$

35. Find distance of the point $(-1, -5, -10)$ from the point of line

$$\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12} \text{ and the plane } x - y + z = 5?$$

Ans: It is clear that $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ has $(2, -1, 2)$ on it.

Place $(2, -1, 2)$ in $x - y + z = 5$

$$2 + 1 + 2 = 5$$

So, it is clear that $(2, -1, 2)$ is the intersection point of $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and $x - y + z = 5$.

Distance between $(2, -1, 2)$ and $(-1, -5, -10)$ is

$$\sqrt{(2 - (-1))^2 + (-1 - (-5))^2 + (2 - (-10))^2} = \sqrt{169} = 13$$

36. Find equation of the line passing through the points $(2, 3, -4)$ and $(1, -1, 3)$ and parallel to X-axis?

Ans: Equation of the plane passing through $(2, 3, -4)$ and $(1, -1, 3)$ and parallel to X-axis is

$$\begin{vmatrix} x-2 & y-3 & z+4 \\ 1 & -1 & 3 \\ 1 & 0 & 0 \end{vmatrix} = 0$$

$$\Rightarrow 3y + z - 5 = 0$$

37. Find the distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$

measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{6}$

Ans: Direction cosines of the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{6}$ are $(2, 3, -6)$

since parallel lines have proportionate direction ratios, so, equation of the line through $P(1, -2, 3)$ and parallel to the given line is

$$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{6} = \lambda \quad \dots\dots\dots (1)$$

Coordinates of any random point on line (1) is $Q(2\lambda + 1, 3\lambda - 2, 3 - 6\lambda)$

If Q lies on the given equation of plane $x - y + z = 5$ then, we have

$$(2\lambda + 1) - (3\lambda - 2) + (3 - 6\lambda) = 5$$

$$\lambda = \frac{1}{7}$$

So, coordinates of the point Q are $Q(\frac{9}{7}, \frac{-11}{7}, \frac{15}{7})$

Therefore, required distance

$$\begin{aligned} PQ &= \sqrt{(\frac{9}{7} - 1)^2 + (\frac{-11}{7} + 2)^2 + (\frac{15}{7} - 3)^2} \\ &= 1 \end{aligned}$$

38. Find the equation of the plane through the intersection of the planes $3x - 4y + 5z = 10$ and $2x + 2y - 3z = 4$ and parallel to the line $x = 2y = 3z$

Ans: The equation of the plane passing through the intersection of the given planes is

$$\begin{aligned} (3x - 4y + 5z - 10) + k(2x + 2y - 3z - 4) &= 0 \\ \Rightarrow 3x - 4y + 5z - 10 + 2kx + 2ky - 3kz - 4k &= 0 \\ \Rightarrow (3 + 2k)x + (-4 + 2k)y + (5 - 3k)z - 10 - 4k &= 0 \quad \dots (1) \end{aligned}$$

The given line is $x = 2y = 3z$

Dividing this equation by 6, we get

$$\frac{x}{6} = \frac{y}{3} = \frac{z}{2}$$

The direction ratios of this line are proportional to 6, 3, 2

So, the normal to the plane is perpendicular to the line whose direction are proportional to 6, 3, 2

$$\begin{aligned} \Rightarrow (3 + 2k)6 + (-4 + 2k)3 + (5 - 3k) &= 0 \\ \Rightarrow 18 + 12k - 12 + 6k + 10 - 6k &= 0 \\ \Rightarrow 12 + 16 &= 0 \\ \Rightarrow k &= \frac{-4}{3} \end{aligned}$$

Substituting this equation in (1) we get

$$\begin{aligned} \Rightarrow (3 + 2(\frac{-4}{3}))6 + (-4 + 2(\frac{-4}{3}))3 + (5 - 3(\frac{-4}{3})) &= 0 \\ \Rightarrow 3x - \frac{8}{3}x - 4y - \frac{8}{3}y + 5z + 4z - 10 + \frac{16}{3} &= 0 \end{aligned}$$

$$\Rightarrow \frac{9x-8x}{3} - \frac{12y+8y}{3} + 9z - \frac{30-16}{3} = 0$$

Multiplying both sides by 3, we get,

$$\Rightarrow x - 20y + 27x - 14 = 0$$

$$\Rightarrow x - 20y + 27x = 14$$

39. Find the distance between the planes $2x + 3y - 4z + 5 = 0$ and $4x + 6y - 8z - 11 = 0$

Ans: The given planes are $2x + 3y - 4z + 5 = 0$ and $4x + 6y - 8z - 11 = 0$.

$$4x + 6y - 8z - 11 = 0 \text{ can be written as } 2x + 3y - 4z - \frac{11}{2} = 0.$$

So these two planes are parallel.

$$\text{The distance between these two planes is } \frac{\left| 5 + \frac{11}{2} \right|}{\sqrt{2^2 + 3^2 + 4^2}} = \frac{21}{2\sqrt{29}}$$

40. Find the equations of the planes parallel to the plane $x - 2y + 2z - 3 = 0$ whose perpendicular distance from the point $(1, 2, 3)$ is 1 unit.

Ans: The equation of the planes parallel to the plane $x - 2y + 2z - 3 = 0$ which are at unit distance from the point $(1, 2, 3)$ is $ax + by + cz + d = 0$. If $(b - d) = k(c - a)$, then the positive value of k is

$$\text{Let plane is } x - 2y + 2z + \lambda = 0$$

$$\text{Distance from } (1, 2, 3) = 1$$

$$\Rightarrow \frac{|\lambda + 3|}{5} = 1$$

$$\Rightarrow \lambda = 0, -6$$

$$\Rightarrow a = 1, b = -2, c = 2, d = -6 \text{ or } 0$$

$$\Rightarrow b - d = 4 \text{ or } -2,$$

$$\Rightarrow c - a = 1$$

$$\Rightarrow k = 4 \text{ or } -2$$

41. Show that the lines $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ intersect each other. Find the point of intersection.

Ans: Let $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7} = u$

And $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5} = v$

General points on the line are

$(3u-1, 5u-3, 7u-5)$ and $(v+2, 3v+4, 5v+6)$

Lines intersect if

$3u-1 = v+2,$

$5u-3 = 3v+4$

$7u-5 = 5v+6$ for some u & v

$3u - v = 3 \quad \dots\dots(1)$

$5u - 3v = 7 \quad \dots\dots(2)$

$7u - 5v = 11 \quad \dots\dots(3)$

Solving equations (1) and (2), we get

$u = \frac{1}{2}, v = \frac{-3}{2}$

Putting u & v in equation (3),

$7(\frac{1}{2}) - 5(\frac{-3}{2}) = 11$

Therefore, lines intersect

Putting value of u & v in general points,

Point of intersection of lines is : $(\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2})$

42. Find the shortest distance between the lines

$\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$

$\vec{r} = 2\hat{i} + 4\hat{j} + 5\hat{k} + \mu(3\hat{i} + 4\hat{j} + 5\hat{k})$

Ans: Given equation of lines are

$\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k}) \quad \dots\dots(1)$

$\vec{r} = 2\hat{i} + 4\hat{j} + 5\hat{k} + \mu(3\hat{i} + 4\hat{j} + 5\hat{k}) \quad \dots\dots(2)$

Position vector and normal vector of line (1) is

$\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$

$\vec{n}_1 = 2\hat{i} + 3\hat{j} + 4\hat{k}$

Position vector and normal vector of line (2) is

$$\vec{c} = 2\hat{i} + 4\hat{j} + 5\hat{k}$$

$$\vec{n}_2 = 3\hat{i} + 4\hat{j} + 5\hat{k}$$

Shortest distance between two skew lines is

$$SD = \frac{\vec{AC} \cdot (\vec{n}_1 \times \vec{n}_2)}{|\vec{n}_1 \times \vec{n}_2|}$$

$$\vec{AC} = \vec{c} - \vec{a}$$

$$\vec{AC} = 2\hat{i} + 4\hat{j} + 5\hat{k} - (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\vec{AC} = 2\hat{i} + 4\hat{j} + 5\hat{k} - \hat{i} - 2\hat{j} - 3\hat{k}$$

$$\vec{AC} = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

$$= \hat{i}(15 - 16) - \hat{j}(10 - 12) + \hat{k}(8 - 9)$$

$$= -\hat{i} + 2\hat{j} - \hat{k}$$

$$|\vec{n}_1 \times \vec{n}_2| = \sqrt{(-1)^2 + 2^2 + (-1)^2}$$

$$= \sqrt{6}$$

Substituting the obtained values in the formula, we get

$$SD = \frac{(\hat{i} + 2\hat{j} + 2\hat{k}) \cdot (-\hat{i} + 2\hat{j} - \hat{k})}{\sqrt{6}}$$

$$SD = \frac{1}{\sqrt{6}}$$

43. Find the distance of the point (-2,3,-4) from the line

$$\frac{x+2}{3} = \frac{2y+3}{4} = \frac{3z+4}{5} \text{ measured parallel to the plane } 4x + 12y - 3z + 1 = 0.$$

Ans: Consider the equation,

$$\frac{x+2}{3} = \frac{2y+3}{4} = \frac{3z+4}{5} = \lambda$$

Therefore, any point on this line is of the form,

$$[3\lambda - 2, \frac{4\lambda - 3}{2}, \frac{5\lambda - 4}{3}]$$

Now, the line from the point $(-2, 3, -4)$ is $3\lambda, \frac{4\lambda - 9}{2}, \frac{5\lambda + 8}{3}$

The equation of the plane is $4x + 12y - 3z + 1 = 0$

Thus, direction ratio of normal is $4, 12, -3$

Therefore,

$$4(3\lambda) + 12(\frac{4\lambda - 9}{2}) - 3(\frac{5\lambda + 8}{3}) = 0$$

$$12\lambda + 24\lambda - 54 - 5\lambda - 8 = 0$$

$$31\lambda = 62$$

$$\lambda = 2$$

Hence, the required coordinates is $(4, \frac{5}{2}, 2)$. Hence, the distance between the coordinates $(4, \frac{5}{2}, 2)$ and $(2, 3, -4)$ is $\frac{17}{2}$ unit.

45. Find the equation of a plane passing through $(-1, 3, 2)$ and parallel to each of the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and $\frac{x+2}{-3} = \frac{y-1}{2} = \frac{z+1}{5}$.

Ans: The directional ratios of the plane is $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ -3 & 2 & 5 \end{vmatrix} = 4\hat{i} - 14\hat{j} + 8\hat{k}$

Let us assume the equation of the plane is $ax + by + cz = d$.

Given that the plane passes through $(-1, 3, 2)$.

So,

$$\Rightarrow d = 4(-1) - 14(3) + 8(2)$$

$$\Rightarrow d = -30$$

The equation of the plane is $-x + 3y + 2z + 30 = 0$.

46. Show that the plane $\vec{r} \cdot (\hat{i} - 3\hat{j} + 3\hat{k}) = 7$ contains the line

$$\vec{r} = (\hat{i} + 3\hat{j} + 3\hat{k}) + \lambda(3\hat{i} + \hat{j}).$$

Ans: If the plane contains a line, then the line must be perpendicular to the normal of the plane.

As,

$$\Rightarrow a_1a_2 + b_1b_2 + c_1c_2 = 0$$

$$\Rightarrow 1(3) - 3(1) + 3(0) = 0$$

So, we can say that the plane $\vec{r} \cdot (\hat{i} - 3\hat{j} + 3\hat{k}) = 7$ contains the line

$$\vec{r} = (\hat{i} + 3\hat{j} + 3\hat{k}) + \lambda(3\hat{i} + \hat{j}).$$

Long Answer Questions

6 Marks

47. Check the coplanarity of the lines

$$\vec{r} = (-3\hat{i} + \hat{j} + 5\hat{k}) + \lambda(-3\hat{i} + \hat{j} + 5\hat{k})$$

$$\vec{r} = (-\hat{i} + 2\hat{j} + 5\hat{k}) + \mu(-\hat{i} + 2\hat{j} + 5\hat{k})$$

If they are coplanar, find the equation of the plane containing the lines.

Ans: Given lines are

$$\vec{r} = (-3\hat{i} + \hat{j} + 5\hat{k}) + \lambda(-3\hat{i} + \hat{j} + 5\hat{k})$$

$$\vec{r} = (-\hat{i} + 2\hat{j} + 5\hat{k}) + \mu(-\hat{i} + 2\hat{j} + 5\hat{k})$$

Now, compare the lines with

$$\vec{r} = \vec{a} + \lambda\vec{b}$$

$$\vec{r} = \vec{c} + \lambda\vec{d}$$

By comparing with them, we get

$$\vec{a} = (-3\hat{i} + \hat{j} + 5\hat{k})$$

$$\vec{b} = (-3\hat{i} + \hat{j} + 5\hat{k})$$

$$\vec{c} = (-\hat{i} + 2\hat{j} + 5\hat{k})$$

$$\vec{d} = (-\hat{i} + 2\hat{j} + 5\hat{k})$$

Now, checking the coplanarity of the lines

$$\vec{c} - \vec{a} = 2\hat{i} + \hat{j} + 0\hat{k}$$

$$(\vec{c} - \vec{a}) \cdot (\vec{b} \times \vec{d})$$

$$\begin{vmatrix} 2 & 1 & 0 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix} \Rightarrow 2(5-10)-1(-15+5)=2(-5)+10=0$$

Hence, they are coplanar.

The equation of the plane containing the lines is

$$(\vec{r} - \vec{a}).(\vec{b} \times \vec{d}) = 0$$

$$\vec{b} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix}$$

$$= \hat{i}(5-10) - \hat{j}(-15+5) + \hat{k}(-6+1)$$

$$= -5\hat{i} + 10\hat{j} - 5\hat{k}$$

$$\vec{r}.(\vec{b} \times \vec{d}) - \vec{a}.(\vec{b} \times \vec{d}) = 0$$

$$\Rightarrow \vec{r}.(-5\hat{i} + 10\hat{j} - 5\hat{k}) - (-3\hat{i} + \hat{j} + 5\hat{k}).(-5\hat{i} + 10\hat{j} - 5\hat{k}) = 0$$

$$\Rightarrow (x\hat{i} + y\hat{j} + z\hat{k}).(-5\hat{i} + 10\hat{j} - 5\hat{k}) - (15 + 10 - 25) = 0$$

$$\Rightarrow -5x + 10y - 5z = 0$$

$$\Rightarrow 5x - 10y + 5z = 0$$

$$\Rightarrow x - 2y + z = 0$$

Therefore, the equation of the plane containing two lines is $x - 2y + z = 0$

48. Find shortest distance between the lines :

$$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7} \text{ and } \frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$$

Ans: Equation of two given lines are

$$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}$$

$$\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$$

First line passes through $(8, -9, 10)$ and has direction ratios proportional to $\langle 3, -16, 7 \rangle$

So, its vector equation is

$$\vec{r} = \vec{a}_2 + \lambda \vec{b}_2$$

Where $\vec{a}_2 = 15\hat{i} + 29\hat{j} + 5\hat{k}$ and $\vec{b}_2 = 3\hat{i} - 8\hat{j} - 5\hat{k}$

$$\vec{a}_2 - \vec{a}_1 = (15\hat{i} + 29\hat{j} + 5\hat{k}) - (8\hat{i} - 9\hat{j} + 10\hat{k}) = 7\hat{i} + 38\hat{j} - 5\hat{k}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix}$$

$$= (80 - 56)\hat{i} - (-15 - 21)\hat{j} + (24 + 48)\hat{k}$$

$$= 24\hat{i} + 36\hat{j} + 72\hat{k}$$

$$= (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)$$

$$= (7\hat{i} + 38\hat{j} - 5\hat{k}) \cdot (24\hat{i} + 36\hat{j} + 72\hat{k})$$

$$= 168 + 1368 - 360$$

$$= 1176$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{(24)^2 + (36)^2 + (72)^2} = \sqrt{7056} = 84$$

$$\text{Shortest distance is equals to } \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right| = \frac{1176}{84} = 14$$

Find the shortest distance between the lines:

$$\vec{r} = (1 - \lambda)\hat{i} + (\lambda - 2)\hat{j} + (3 - 2\lambda)\hat{k}$$

$$\vec{r} = (\mu + 1)\hat{i} + (2\mu - 1)\hat{j} + (2\mu + 1)\hat{k}$$

50. A variable plane is at a constant distance $3p$ from the origin and meets the coordinate axes in A, B and C. If the centroid of ΔABC is (α, β, γ) , then show that $\alpha^{-2} + \beta^{-2} + \gamma^{-2} = p^{-2}$

Ans: Let the equation of the plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ where a, b, c are intercepts of plane on x, y, z axis respectively.

The distance from the origin to plane will be

$$\frac{1}{\sqrt{\alpha^{-2} + \beta^{-2} + \gamma^{-2}}} = 3p$$

$$\Rightarrow \alpha^{-2} + \beta^{-2} + \gamma^{-2} = \frac{1}{9p^2}$$

The centroid of the triangle ABC is $(\frac{a}{3}, \frac{b}{3}, \frac{c}{3})$ and let it be (α, β, γ)

So, we have $a = 3\alpha, b = 3\beta, c = 3\gamma$

$$\Rightarrow \frac{1}{9\alpha^2} + \frac{1}{9\beta^2} + \frac{1}{9\gamma^2} = \frac{1}{9p^2}$$

$$\Rightarrow \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{1}{p^2}$$

Hence proved.

51. A vector $\vec{n}$ of magnitude 8 units is inclined to x-axis at 45° , y axis at 60° and an acute angle with z-axis. If a plane passes through a point $(\sqrt{2}, -1, 1)$ and is normal to $\vec{n}$, find its equation in vector form?

Ans: let γ be the angle made by $\vec{n}$ with z-axis, then the directions of cosines of $\vec{n}$, are

$$l = \cos 45^\circ = \frac{1}{\sqrt{2}}, m = \cos 60^\circ = \frac{1}{2}, n = \cos \gamma$$

$$\therefore l^2 + m^2 + n^2 = 1$$

$$\Rightarrow \frac{1}{2} + \frac{1}{4} + n^2 = 1$$

$$\Rightarrow n^2 = \frac{1}{4}$$

$$\Rightarrow n = \frac{1}{2}$$

(neglecting $n = -\frac{1}{2}$ as γ is acute, $\therefore n > 0$)

We have $|\vec{n}| = 8$

$$\therefore \vec{n} = |\vec{n}|(l\hat{i} + m\hat{j} + n\hat{k})$$

$$\Rightarrow \vec{n} = 8\left(\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{2}\hat{j} + \frac{1}{2}\hat{k}\right)$$

$$\Rightarrow 4\sqrt{2}\hat{i} + 4\hat{j} + 4\hat{k}$$

The required plane passes through the point $(\sqrt{2}, -1, 1)$ having position vector

$$\vec{a} = \sqrt{2}\hat{i} - \hat{j} + \hat{k}$$

So, its vector equation is: $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$

$$\Rightarrow \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$$

$$\Rightarrow \vec{r} \cdot (4\sqrt{2}\hat{i} + 4\hat{j} + 4\hat{k}) = (\sqrt{2}\hat{i} - \hat{j} + \hat{k}) \cdot (4\sqrt{2}\hat{i} + 4\hat{j} + 4\hat{k})$$

$$\Rightarrow \vec{r} \cdot (4\sqrt{2}\hat{i} + 4\hat{j} + 4\hat{k})$$

$$\Rightarrow \vec{r} \cdot (\sqrt{2}\hat{i} + \hat{j} + \hat{k}) = 2$$

52. Find the foot of the perpendicular from the point $2\hat{i} - \hat{j} + 5\hat{k}$ on the line $\vec{r} = (11\hat{i} - 2\hat{j} - 8\hat{k}) + \lambda(10\hat{i} - 4\hat{j} - 11\hat{k})$. Also find the length of the perpendicular.

Ans: Let L be the foot of the perpendicular drawn from $p(2\hat{i} - \hat{j} + 5\hat{k})$ on the line $\vec{r} = (11\hat{i} - 2\hat{j} - 8\hat{k}) + \lambda(10\hat{i} - 4\hat{j} - 11\hat{k})$

Let the position vector of L be

$$11\hat{i} - 2\hat{j} - 8\hat{k} + \lambda(10\hat{i} - 4\hat{j} - 11\hat{k}) = (11 + 10\lambda)\hat{i} + (-2 - 4\lambda)\hat{j} + (-8 - 11\lambda)\hat{k}$$

Then,

$\vec{PL}$ is equal to the difference between position vector of L and position vector of P

$$(11 + 10\lambda)\hat{i} + (-2 - 4\lambda)\hat{j} + (-8 - 11\lambda)\hat{k} - (2\hat{i} - \hat{j} + 5\hat{k}) =$$

$$(9 + 10\lambda)\hat{i} + (-1 - 4\lambda)\hat{j} + (-13 - 11\lambda)\hat{k}$$

Since PL is perpendicular to the given line and the given line is parallel to $b = 10\hat{i} - 4\hat{j} - 11\hat{k}$

Therefore

$$\Rightarrow \vec{PL} \cdot \vec{b} = 0$$

$$\Rightarrow [(9 + 10\lambda)\hat{i} + (-1 - 4\lambda)\hat{j} + (-13 - 11\lambda)\hat{k}] \cdot (10\hat{i} - 4\hat{j} - 11\hat{k}) = 0$$

$$\Rightarrow 10(9 + 10\lambda) - 4(-1 - 4\lambda) - 11(-13 - 11\lambda) = 0$$

$$\Rightarrow 237\lambda = -237$$

$$\Rightarrow \lambda = -1$$

Putting the value of λ , we obtain the position vector of L as $\hat{i} + 2\hat{j} + 3\hat{k}$

Now,

$$\vec{PL} = (\hat{i} + 2\hat{j} + 3\hat{k}) - (2\hat{i} - \hat{j} + 5\hat{k}) = -\hat{i} + 3\hat{j} - 2\hat{k}$$

Hence, the length of the perpendicular from P on the given line is

$$|\vec{PL}| = \sqrt{1+9+4} = \sqrt{14}$$

53. A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube. Prove that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}$

Ans: Take O as corner

OA, OB, OC are 3 edges through the axes.

Let OA=OB=OC=a

Coordinates of O = (0, 0, 0)

A(a, 0, 0) B(0, a, 0) C(0, 0, a)

P(a, a, 0) L(0, a, a) M(a, 0, a) N(a, a, 0)

The four diagonals OP, AL, BM, CN

Direction cosines of OP : a-0, a-0, a-0 = a, a, a = 1, 1, 1

Direction cosines of AL : 0-a, a-0, a-0 = -a, a, a = -1, 1, 1

Direction cosines of BM : a-0, 0-a, a-0 = a, -a, a = 1, -1, 1

Direction cosines of CN : a-0, a-0, 0-a = a, a, -a = 1, 1, -1

Therefore,

Direction cosines of OP are $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

Direction cosines of AL are $\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

Direction cosines of BM are $\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

Direction cosines of CN are $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}$

Let l, m, n are the direction cosines of line and line makes angle α with OP

$$\cos \alpha = l\left(\frac{1}{\sqrt{3}}\right) + m\left(\frac{1}{\sqrt{3}}\right) + n\left(\frac{1}{\sqrt{3}}\right) = \frac{l+m+n}{\sqrt{3}}$$

Similarly,

$$\cos \beta = \frac{-l+m+n}{\sqrt{3}}$$

$$\cos \gamma = \frac{l+m-n}{\sqrt{3}}$$

$$\cos \delta = \frac{l-m+n}{\sqrt{3}}$$

Squaring and adding all the four, we get

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{1}{3}(l+m+n)^2 + (-l+m+n)^2 + (l-m+n)^2 + (l+m-n)^2$$

$$= \frac{1}{3}(4l^2 + 4m^2 + 4n^2)$$

$$= \frac{4}{3}(l^2 + m^2 + n^2)$$

$$[\because l^2 + m^2 + n^2 = 1]$$

$$= \frac{4}{3}$$

$$\therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}$$

54. Find the equation of the plane passing through the intersection of planes $2x + 3y - z = -1$ and $x + y - 2z + 3 = 0$ and perpendicular to the plane $3x - y - 2z = 4$. Also find the inclination of the plane with xy -plane.

Ans: Given equation of the planes are $2x + 3y - z = -1$ and $x + y - 2z + 3 = 0$

As we know that

The equation of the plane passes through the line of intersection of the planes

$$a_1x + b_1y + c_1z + d_1 = 0 \quad \text{and} \quad a_2x + b_2y + c_2z + d_2 = 0 \quad \text{is}$$

$$a_1x + b_1y + c_1z + d_1 + \lambda(a_2x + b_2y + c_2z + d_2) = 0$$

So, the required equation of the plane is

$$2x + 3y - z + 1 + \lambda(x + y - 2z + 3) = 0$$

$$(2 + \lambda)x + (3 + \lambda)y + (-1 - 2\lambda)z + 1 + 3\lambda = 0 \quad \dots\dots\dots(1)$$

This plane is perpendicular to $3x - y - 2z - 4 = 0$

Substituting the λ value in (1)

$$(2 - \frac{5}{6})x + (3 - \frac{5}{6})y + (-1 - 2(-\frac{5}{6}))z + 1 + 3(-\frac{5}{6}) = 0$$

$$\Rightarrow 7x + 13y + 4z - 9 = 0$$

Therefore, the equation of required plane is $7x + 13y + 4z - 9 = 0$.